

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
4	(a)		$T = 0.60$ [s] or $f = 1.67$ [Hz] in working or by implication (1) Sinusoid of correct frequency and amplitude drawn from $t = 0$ to $t = 1$ s (1) Reasonable graph: Correct phase ($-\sin \omega t$) (1)		3		3	2	
	(b)	(i)	$1\,500$ [nm] $\sin 24.9^\circ = \lambda$ or by implication (1) $\lambda = 632$ nm (1)	1	1		2	2	
		(ii)	Diagram of right-angled triangle (by eye) with 57.4° (accept θ) marked in a correct position [i.e. wavefront from bottom slit perpendicular to top direction] (1) Either 2λ (or path difference) marked on diagram or statement that path difference [for light from adjacent slits] = 2λ (1) 57.4° (or θ) = $\sin^{-1} \frac{2\lambda}{d}$ or equivalent [e.g. $\sin^{-1}(2 \sin 24.9^\circ)$] (1) [Last mark free-standing]	1 1	1		3		
			Question 4 total	3	5	0	8	4	0